$$
\begin{array}{l}
\quad [k \in Keys \mapsto \\
\quad\quad \text{IF } tx\_buffer[t][k] \neq \langle\rangle \\
\quad\quad\quad \text{THEN } store[k] \cup \\
\quad\quad\quad\quad \{[val \mapsto tx\_buffer[t][k][1], \\
\quad\quad\quad\quad\quad cts \mapsto shard\_ct', \\
\quad\quad\quad\quad\quad writer \mapsto t \\
\quad\quad\quad\quad] \\
\quad\quad\quad\quad \} \\
\quad\quad\quad \text{ELSE } store[k] \\
\quad] \\
\land\ client\_dep' = \\
\quad \text{IF } shard\_ct' > client\_dep[Workload[t].session] \\
\quad\quad \text{THEN } [client\_dep \text{ EXCEPT } ![Workload[t].session] = shard\_ct'] \\
\quad\quad \text{ELSE } client\_dep \\
\land\ client\_last\_commit' = \\
\quad [client\_last\_commit \text{ EXCEPT} \\
\quad ![Workload[t].session] = \\
\quad shard\_ct'] \\
\land\ \text{UNCHANGED } \langle router\_ct, \\
\quad\quad tx\_snapshot, \\
\quad\quad tx\_checkset, \\
\quad\quad tx\_buffer, \\
\quad\quad pc, \\
\quad\quad op\_seq \\
\quad \rangle
\end{array}
$$

$$
\begin{array}{l}
Next \triangleq \\
\quad (\exists\, t \in TxnIds : StartTx(t) \lor ExecuteOp(t) \lor CommitTx(t)) \lor \\
\quad (\forall\, t \in TxnIds : \\
\quad\quad tx\_status[t] \in \{\text{“Committed”}, \text{“Aborted”}\} \land \text{UNCHANGED } all\_vars \\
\quad)
\end{array}
$$

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2. AXIOMATIC FRAMEWORK & MAPPING

$$
CommittedTxns \triangleq \{t \in AllTxns : tx\_status[t] = \text{“Committed”}\}
$$

$$
\begin{array}{l}
Derived\_AR \triangleq \\
\quad \{\langle t1, t2\rangle \in CommittedTxns \times CommittedTxns : \\
\quad\quad tx\_commit\_time[t1] < tx\_commit\_time[t2] \\
\quad \}
\end{array}
$$

$$
\begin{array}{l}
Derived\_VIS \triangleq \\
\quad \{\langle t1, t2\rangle \in CommittedTxns \times CommittedTxns : \\
\quad\quad \lor \land\ t2 \neq InitTx \\
\quad\quad\quad \land\ Workload[t2].level \in \{\text{“RA”}, \text{“CC”}, \text{“PC”}, \text{“PSI”}, \text{“SI”}\}
\end{array}
$$